

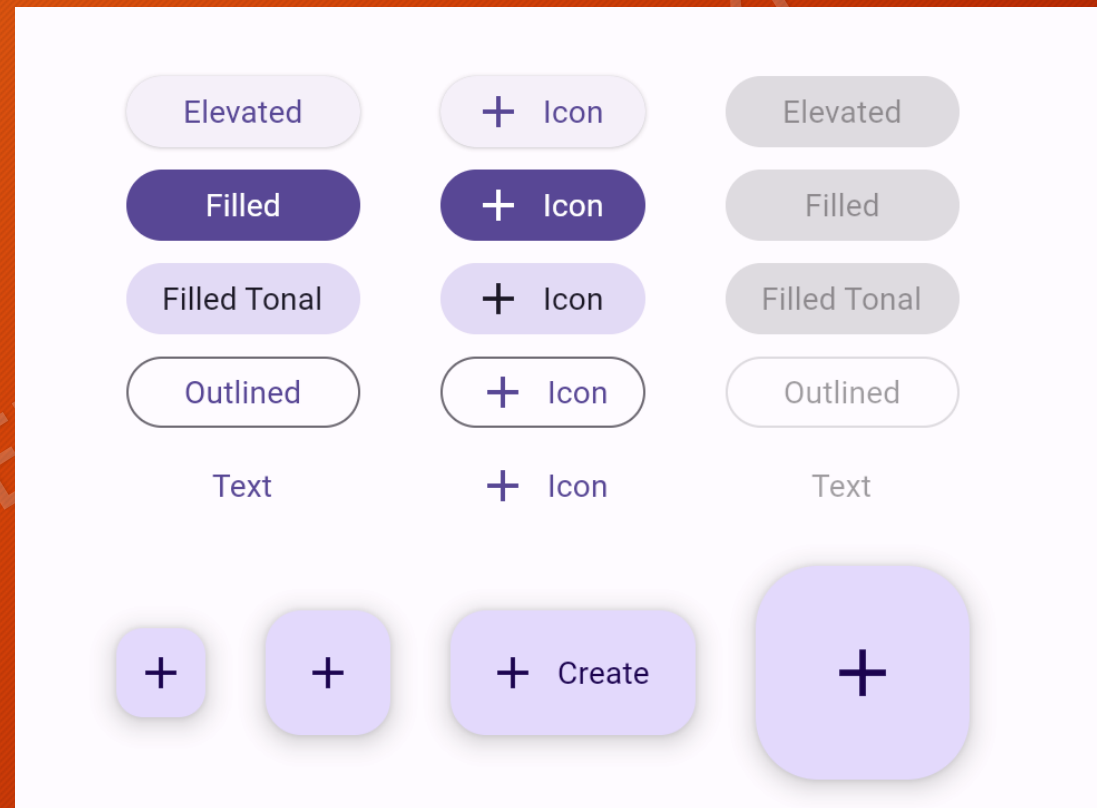
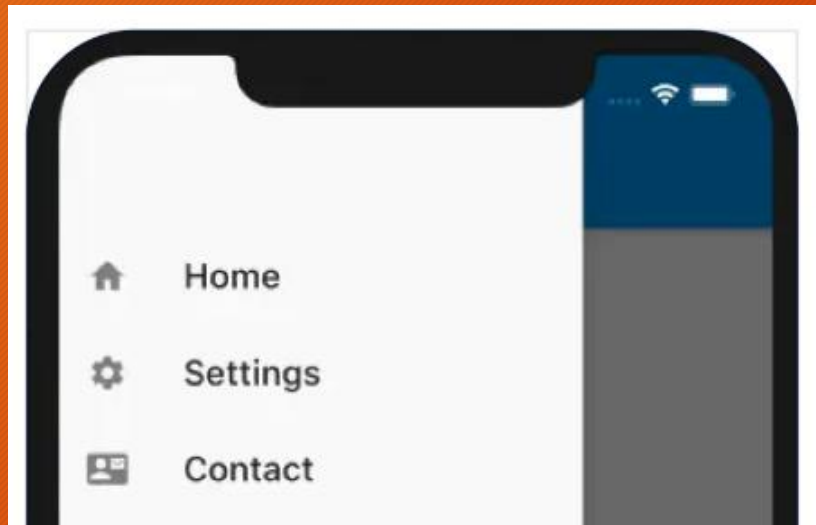
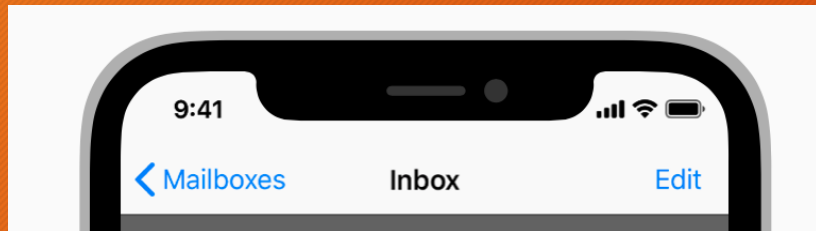
Mobile Application Development with Flutter

Lecture 13: Flutter Project Structure

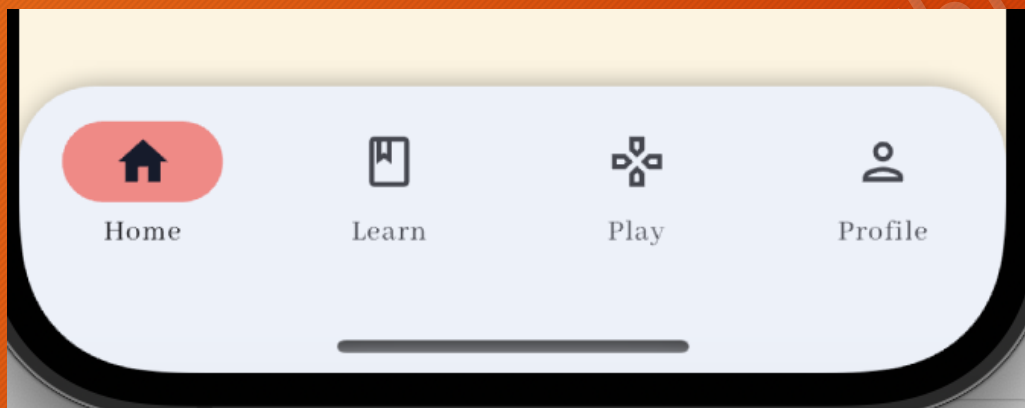
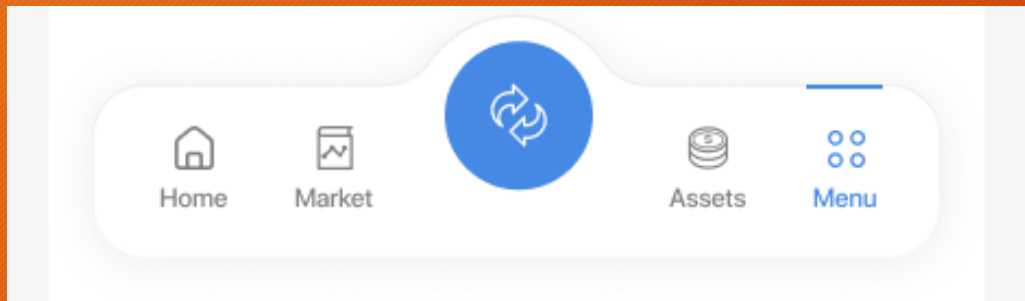
Widgets

- Widgets are the primary component of any flutter application
- It acts as a UI for the user to interact with the application
- Everything on the screen, including text, buttons, and even the layout structure (like rows and columns), is a widget
- Widgets are organized into a hierarchical structure called the widget tree
- There are two main categories of widgets
 - Stateless Widget
 - Stateful Widget

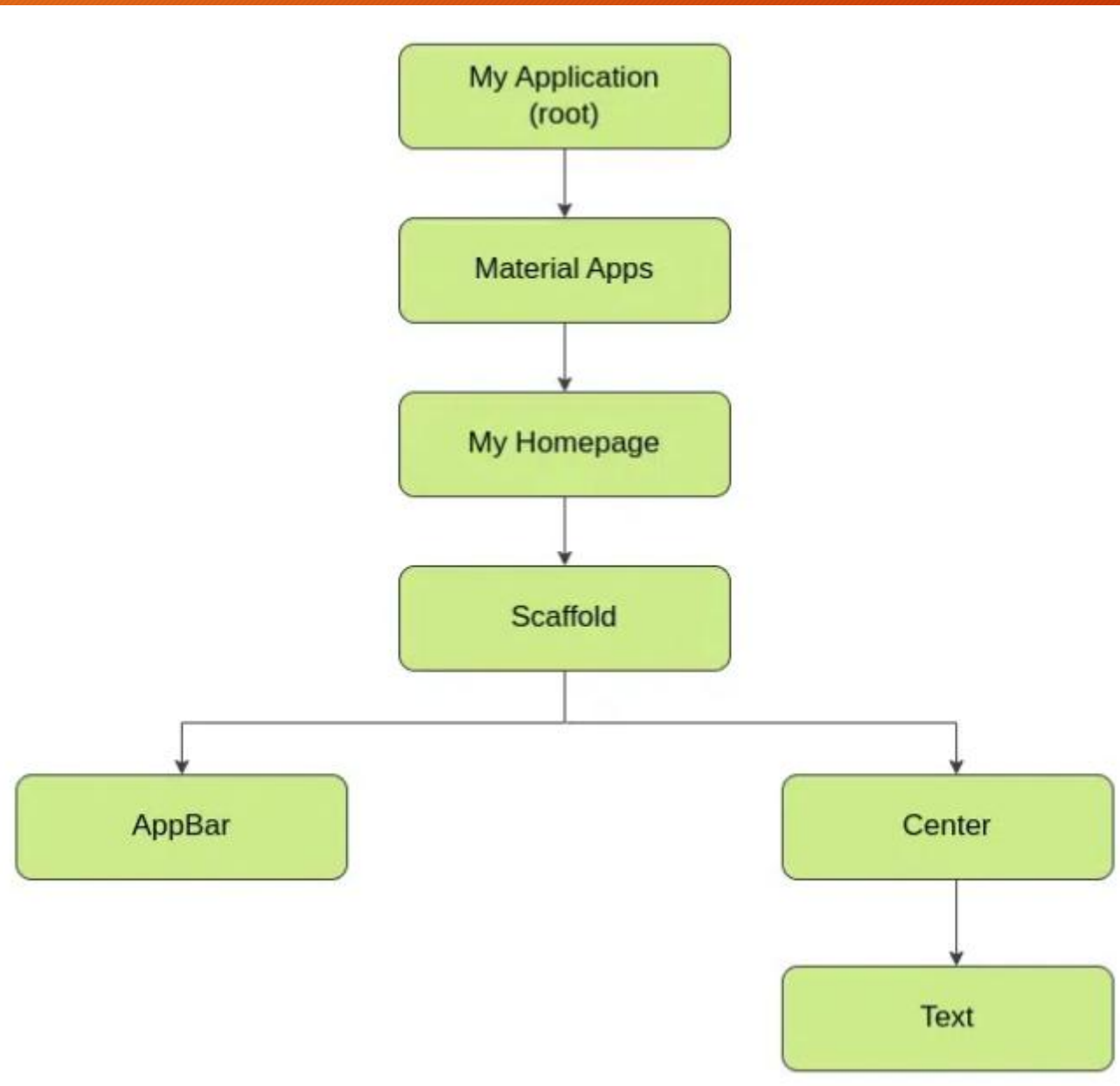
Different Widgets



Different Widgets



Widget Tree



Understand Existing Code

- The main() function (Entry point of Flutter app)
 - Every Dart program starts from main()
 - From here we call runApp() to attach your widget to the screen
- runApp() (Launch our widget on the screen)
 - runApp() takes a **Widget** and displays it as the root of the app
- What happens when runApp() runs?
 - Flutter creates the UI from the given widget
 - That widget becomes the **root** of the widget tree
 - Everything on screen is built from that root widget

MaterialApp (The app wrapper for Material Design)

- MaterialApp is usually our **top-level widget**
- Why we use MaterialApp?
 - Material Design styling
 - Navigation system (routes)
 - Theme (colors, fonts)
 - Title
 - Debug banner control
 - App-wide settings

Prepared by Ehtisham Rasheed

Scaffold (The screen layout structure)

- Scaffold is like a ready-made page layout
- It gives us:
 - App bar
 - Body
 - Floating action button
 - Drawer
 - Bottom navigation bar
 - Snackbars
 - etc

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